

DEPARTMENT OF MATHEMATICS
UDAI PRATAP COLLEGE, VARANASI

Syllabus as per NEP-2020 guidelines
FOR FIRST THREE YEARS OF HIGHER EDUCATION (UG)



For
B.A./ B.Sc.

MATHEMATICS

(EFFECTIVE FROM THE SESSION 2022-23)

SEMESTER WISE TITLES OF THE PAPER IN UG MATHEMATICS COURSE					
YEAR	SEMESTER	COURSE CODE	PAPER TITLE	THEORY/PRACTICAL	CREDIT
CERTIFICATE COURSE IN APPLIED MATHEMATICS					
FIRST YEAR	I	B030101T	Differential Calculus & Integral Calculus	THEORY	4
		B030102P	PRACTICAL	PRACTICAL	2
	II	B030201T	Matrices and Differential Equations & Geometry	THEORY	6
DIPLOMA IN MATHEMATICS					
SECOND YEAR	III	B030301T	Algebra & Mathematical Methods	THEORY	6
	IV	B030401T	Differential Equation & Mechanics	THEORY	6
DEGREE IN MATHEMATICS					
THIRD YEAR	V	B030501T	Group and Ring Theory & Linear Algebra	THEORY	5
		B030502T	Any One of The Following (i) Number Theory & Game Theory (ii) Graph Theory & Discrete Mathematics (iii) Differential Geometry & Tensor Analysis	THEORY	5
		B030503R	Research Project/Dissertation	PROJECT	
	VI	B030601T	Metric Space & Complex Analysis	THEORY	4
		B030602T	Numerical Analysis & Operations Research	THEORY	4
		B030603P	PRACTICAL	PRACTICAL	2
		B030604R	Research Project/Dissertation	PROJECT	

PROPOSED STRUCTURE OF UG MATHEMATICS SYLLABUS AS PER NEP 2020 GUIDELINES

GENERAL OVERVIEW

B.A./B.Sc. I										
PROGRAMME	YEAR	SEMESTER (15 Weeks)	PAPER	CREDIT	PERIODS Per Week	PERIODS (HOURS) Per Semester	PAPER TITLE	UNIT (Periods Per Semester)	PREREQUISITE	ELECTIVE (For Other Faculty)
CERTIFICATE COURSE IN APPLIED MATHEMATICS	FIRST YEAR	SEMESTER – I	Paper-1	4	4	4x 15= 60	Differential Calculus & Integral Calculus Part A: Differential Calculus Part B: Integral Calculus	Part A Unit I (9) Unit II (7) Unit III (7) Unit IV (7) Part B Unit V (9) Unit VI (7) Unit VII (7) Unit VIII (7)	Mathematics in 12 th	Engg. and Tech. (UG), Chemistry/Biochemistry/ Life Sciences(UG), Economics(UG/PG), Commerce(UG), BBA/BCA, B.Sc.(C.S.)
			Paper-II Practical	2	2 Lab Periods(2 Hours Each)	2x2x 15= 60	Practical (PRACTICALS TO BE DONE USING MATHEMATICA/MATLAB /MAPLE/SCILAB/MAXIMA /OCATVE ETC.)		Mathematics in 12 th	Engg. and Tech. (UG), B.Sc.(C.S.)
		SEMESTER – II	Paper-1	6	6	6 x 15= 90	Matrices and Differential Equations & Geometry Part A: Matrices and Differential Equations Part B: Geometry	Part A Unit I (12) Unit II (11) Unit III (11) Unit IV (11) Part B Unit V (12) Unit VI (11) Unit VII (11) Unit VIII (11)	Mathematics in 12 th	Engg. and Tech. (UG), B.Sc.(C.S.)

B.A./B.Sc. II										
PROGRAMME	YEAR	SEMESTER (15 Weeks)	PAPER	CREDIT	PERIODS Per Week	PERIODS (HOURS) Per Semester	PAPER TITLE	UNIT (Periods Per Semester)	PREREQUISITE	ELECTIVE (For Other Faculty)
DIPLOMA IN MATHEMATICS	SECOND YEAR	SEMESTER –III	Paper-1	6	6	6 x 15= 90	Algebra & Mathematical Methods Part A: Algebra Part B: Mathematical Methods	Part A Unit I (12) Unit II (11) Unit III (11) Unit IV (11) Part B Unit V(12) Unit VI (11) Unit VII (11) Unit VIII (11)	Certificate Course in Applied Mathematics	Engg. and Tech. (UG), B.Sc.(C.S.)
		SEMESTER – IV	Paper-1	6	6	6 x 15= 90	Differential Equation & Mechanics Part A: Differential Equation Part B: Mechanics	Part A Unit I (12) Unit II (11) Unit III (11) Unit IV (11) Part B Unit V(12) Unit VI (11) Unit VII (11) Unit VIII (11)	Certificate Course in Applied Mathematics	Engg. and Tech. (UG), Economics(UG/PG), B.Sc.(C.S.) Engineering and Technology (UG), Science (Physics-UG)

B.A./B.Sc. III										
PROGRAMME	YEAR	SEMESTER (15 Weeks)	PAPER	CREDIT	PERIODS Per Week	PERIODS (HOURS) Per Semester	PAPER TITLE	UNIT (Periods Per Semester)	PREREQUISITE	ELECTIVE (For Other Faculty)
DEGREE IN MATHE THIRD YEAR		SEMESTER – V	Paper-1	5	5	5x 15= 75	Group and Ring Theory & Linear Algebra Part A: Group and Ring Theory Part B: Linear Algebra	Part A Unit I (10) Unit II(10) Unit III(9) Unit IV (9) Part B Unit V(10) Unit VI(9) Unit VII (9) Unit VIII (9)	Certificate Course in Applied Mathematics	Engg. and Tech. (UG), Economics(UG/PG), B.Sc.(C.S.)
			Paper-2	5	5	5x 15= 75	(i) Number Theory & Game Theory Part A: Number Theory Part B: Game Theory	Part A Unit I(10) Unit II(9) Unit III(9) Unit IV (9) Part B Unit V(10) Unit VI(10) Unit VII(9) Unit VIII (9)	Diploma in Mathematics	Engg. and Tech.(UG), BCA, B.Sc.(C.S.)
							(ii) Graph Theory & Discrete Mathematics Part A: Graph Theory Part B: Discrete Mathematics	Part A Unit I(10) Unit II(9) Unit III(9) Unit IV (9) Part B Unit V(10) Unit VI(10) Unit VII(9) Unit VIII (9)	Diploma in Mathematics	Engg. and Tech. (UG), B.Sc.(C.S.)
							(iii) Differential Geometry & TensorAnalysis Part A: Differential Geometry Part B: Tensor Analysis	Part A Unit I(10) Unit II(9) Unit III(9) Unit IV (9) Part B Unit V(10) Unit VI(10) Unit VII(9) Unit VIII (9)	Diploma in Mathematics	Engg. and Tech. (UG), B.Sc.(C.S.)

		SEMESTER – VI	Paper-1	4	4	4 x 15= 60	Metric Space & Complex Analysis Part A: Metric Space Part B: Complex Analysis	Part A Unit I (8) Unit II (8) Unit III (7) Unit IV (7) Part B Unit V (8) Unit VI (8) Unit VII (7) Unit VIII (7)	Diploma in Mathematics	Engg. and Tech. (UG), B.Sc.(C.S.)
			Paper-2	4	4	4x 15= 60	Numerical Analysis & Operations Research Part A: Numerical Analysis Part B: Operations Research	Part A Unit I (8) Unit II (8) Unit III (7) Unit IV (7) Part B Unit V (8) Unit VI (8) Unit VII (7) Unit VIII (7)	Diploma in Mathematics	Engg. and Tech. (UG), Economics(UG/PG), BBA/BCA, B.Sc.(C.S.)
			Paper-III Practical	2	2 Lab Periods(2 Hours Each)	2x2x 15= 60	Practical (Practicals to be done using Mathematica/ MATLAB/Maple/Sci lab/Maxima/ Octave etc.)		Diploma in Mathematics	Engg. and Tech. (UG), B.Sc.(C.S.)
Programme Outcome/ Programme Specific Outcome										
Programme Outcome: PO1: It is to give foundation knowledge for the students to understand basics of mathematics including applied aspect for the same. PO2: It is to develop enhanced quantitative skills and pursuing higher mathematics and research as well. PO3: Students will be able to develop solution oriented approach towards various issues related to their environment. PO4: Students will become employable in various govt. and private sectors PO5: Scientific temper in general and mathematical temper in particular will be developed in students. Programme Specific Outcome: PSO1: Student should be able to possess recall basic idea about mathematics which can be displayed by them. PSO2: Student should have adequate exposure to many aspects of mathematical sciences. PSO3: Student is equipped with mathematical modeling ability, critical mathematical thinking, and problem solving skills etc. PSO4: Student should be able to apply their skills and knowledge in various fields of studies including, science, engineering, commerce and management etc.										

B.A./B.Sc. I (MATHEMATICS)

Detailed Syllabus For

CERTIFICATE COURSE

IN

APPLIED MATHEMATICS

B.A./B.Sc. I (SEMESTER-I) PAPER-I Differential Calculus & Integral Calculus

Programme: Certificate		Year: First	Semester: First
Class: B.A./B.Sc.			
Subject: Mathematics			
Course Code: B030101T		Course Title: Differential Calculus & Integral Calculus	
Course outcomes:			
CO1: The programme outcome is to give foundation knowledge for the students to understand basics of mathematics including applied aspect for developing enhanced quantitative skills and pursuing higher mathematics and research as well.			
CO2: By the time students complete the course they will have wide ranging application of the subject and have the knowledge of real valued functions such as sequence and series. They will also be able to know about convergence of sequence and series. Also, they have knowledge about curvature, envelope and evolutes and trace curve in polar, Cartesian as well as parametric curves.			
CO3: The main objective of the course is to equip the student with necessary analytic and technical skills. By applying the principles of integral he learns to solve a variety of practical problems in science and engineering.			
CO4: The student is equipped with standard concepts and tools at an intermediate to advance level that will serve him well towards taking more advance level course in mathematics.			
Credits: 4		Core Compulsory / Elective	
Max. Marks: 25+75		Min. Passing Marks:	
Total No. of Lectures-Tutorials-Practical (in hours per week): L-T-P: 4-0-0			
Part- A			
Differential Calculus			
Unit	Topics		No. of Lectures
I	Introduction to Indian ancient Mathematics and Mathematicians should be included under Continuous Internal Evaluation (CIE). Definition of a sequence, theorems on limits of sequences, bounded and monotonic sequences, Cauchy's convergence criterion, Cauchy sequence, limit superior and limit inferior of a sequence, subsequence, Series of non-negative terms, convergence and divergence, Comparison tests, Cauchy's integral test, Ratio tests, Root test, Raabe's logarithmic test, de Morgan and Bertrand's tests, alternating series, Leibnitz's theorem, absolute and conditional convergence.		9
II	Limit, continuity and differentiability of function of single variable, Cauchy’s definition, Heine’s definition, equivalence of definition of Cauchy and Heine, Uniform continuity, Borel’s theorem, boundedness theorem, Bolzano’s theorem, Intermediate value theorem, Extreme value theorem, Darboux's intermediate value theorem for derivatives, Chain rule, indeterminate forms.		7
III	Rolle’s theorem, Lagrange and Cauchy Mean value theorems, mean value theorems of higher order, Taylor's theorem with various forms of remainders, Successive differentiation, Leibnitz theorem, Maclaurin’s and Taylor’s series, Partial differentiation, Euler’s theorem on homogeneous function.		7
IV	Tangent and normals, Asymptotes, Curvature, Tests for concavity and convexity, Points of inflexion, Multiple points, Parametric representation of curves and tracing of parametric curves, Tracing of curves in Cartesian and Polar forms.		7

Part-B Integral Calculus		
Unit	Topics	No. of Lectures
V	Definite integrals as limit of the sum, Riemann integral, Integrability of continuous and monotonic functions, Fundamental theorem of integral calculus, Mean value theorems of integral calculus, Differentiation under the sign of Integration.	9
VI	Improper integrals, their classification and convergence, Comparison test, μ -test, Abel's test, Dirichlet's test, quotient test, Beta and Gamma functions.	7
VII	Rectification, Volumes and Surfaces of Solid of revolution, Pappus theorem, Multiple integrals, change of order of double integration, Dirichlet's theorem, Liouville's theorem for multiple integrals.	7
VIII	Vector Differentiation, Gradient, Divergence and Curl, Normal on a surface, Directional Derivative, Vector Integration, Theorems of Gauss, Green, Stokes and related problems.	7
Suggested Readings (Part- A Differential Calculus): 1. R.G. Bartle & D.R. Sherbert, Introduction to Real Analysis, John Wiley & Sons 2. T.M. Apostol, Calculus Vol. I, John Wiley & Sons Inc. 3. S. Balachandra Rao & C. K. Shantha, Differential Calculus, New Age Publication. 4. H. Anton, I. Birens and S. Davis, Calculus, John Wiley and Sons, Inc.,2002. 5. G.B. Thomas and R.L. Finney, Calculus, Pearson Education, 2007. 6. Ajit Kumar and S. Kumaresan, A Basic Course in Real Analysis, CRC Press, 2019 7. Suggestive digital platforms web links: NPTEL/SWAYAM/MOOCs Suggested Readings (Part-B Integral Calculus): 1. T.M. Apostol, Calculus Vol. II, John Wiley Publication 2. Shanti Narayan & Dr. P.K. Mittal, Integral Calculus, S.Chand 3. Erwin Kreyszig, Advanced Engineering Mathematics, John Wiley & Sons. 4. M R Spiegel, S Lipschutz,, Schaum's Outline of Vector Analysis, Schaum's Outlines 5. Suggestive digital platforms web links:NPTEL/SWAYAM/MOOCs		
This course can be opted as an elective by the students of following subjects: Engg.and Tech.(UG), Chemistry/Biochemistry/Life Sciences(UG), Economics(UG/PG), Commerce(UG), BBA/BCA, B.Sc.(C.S.)		
Suggested Continuous Evaluation Methods: Max. Marks: 25		
SN	Assessment Type	Max. Marks
1	Class Tests	10
2	Online Quizzes/ Objective Tests	5
3	Presentation	5
4	Assignment (Introduction to Indian ancient Mathematics and Mathematicians).	5
Course prerequisites: To study this course, a student must have subject Mathematics in class 12 th		
Suggested equivalent online courses:		
Further Suggestions:		

B.A./B.Sc. I (SEMESTER-I) Paper-II Practical

Programme: Certificate		Year: First	Semester: First
Class: B.A./B.Sc.			
Subject: Mathematics			
Course Code: B030102P		Course Title: Practical	
Course outcomes:			
CO1: The main objective of the course is to equip the student to plot the different graph and solve the different types of equations by plotting the graph using different computer software such as Mathematica /MATLAB /Maple /Scilab/Maxima/Octaveetc.			
CO2. After completion of this course student would be able to know the convergence of sequences through plotting, verify Bolzano-Weierstrass theorem through plotting the sequence, Cauchy’s root test by plotting n^{th} roots and Ratio test by plotting the ratio of n^{th} and $(n + 1)^{th}$ term.			
CO3. Student would be able to plot Complex numbers and their representations, Operations like addition, substraction, Multiplication, Division, Modulus and Graphical representation of polarform.			
CO4: Student would be able to perform following task of matrix as Addition, Multiplication, Inverse, Transpose, Determinant, Rank,Eigenvectors, Eigenvalues, Characteristic equation and verification of the Cayley-Hamilton theorem, Solving the systems of linear equations.			
Credits: 2		Core Compulsory / Elective	
Max. Marks: 25+75		Min. Passing Marks:	
Total No. of Lectures-Tutorials-Practical (in hours per week): L-T-P: 0-0-4			
Unit	Topics		No. of Lectures
	Practical / Lab work to be performed in Computer Lab. List of the practicals to be done using Mathematica /MATLAB /Maple /Scilab/Maxima/ Octave etc. (1) Plotting the graphs of the following functions: (i) ax (ii) $[x]$(greatest integerfunction) (iii) $x^{2n}; n \in \mathbb{N}$ (iv) $x^{2n-1}; n \in \mathbb{N}$ (v) $\frac{1}{x^{2n-1}}; n \in \mathbb{N}$ (vi) $\frac{1}{x^{2n}}; n \in \mathbb{N}$ (vii)$\sqrt{ax + b}, ax + b , c \pm ax + b$ (viii) $\frac{ x }{x}, \sin\left(\frac{1}{x}\right), x \sin\left(\frac{1}{x}\right), e^x, e^{-x}$for $x \neq 0$. (ix) $e^{ax+b}, \log(ax + b), \sin(ax + b), \cos(ax + b), \sin(ax + b) , \cos(ax + b)$ Observe and discuss the effect of changes in the real constants a and b on the graphs. (2) By plotting the graph find the solution of theequation $x = e^x, x^2 + 1 = e^x, 1 - x^2 = e^x, x = \log_{10}(x), \cos(x) = x, \sin(x) = x, \cos(y) = \cos(x), \sin(y) = \sin(x)$ etc. (3) Plotting the graphs of polynomial of degree 2,3,4and 5, and their first and secondderivatives.		

	(4) Sketching parametric curves, e.g., Trochoid, Cycloid, Epicycloid and Hypocycloidetc. (5) Tracing of conic in Cartesian coordinates. (6) Graph of circular and hyperbolic functions. (7) Obtaining surface of revolution of curves. (8) Complex numbers and their representations, Operations like addition, Multiplication, Division, Modulus. Graphical representation of polarform. (9) Find numbers between two real numbers and plotting of finite and infinite subset of $\mathbb{R}$. (10) Matrix Operations: Addition, Multiplication, Inverse, Transpose, Determinant, Rank, Eigenvectors, Eigenvalues, Characteristic equation and verification of the Cayley-Hamilton theorem, Solving the systems of linear equations. (11) Study the convergence of sequences through plotting. (12)Verify Bolzano-Weierstrass theorem through plotting of sequences and hence identify convergent subsequences from the plot. (13)Study the convergence/divergence of infinite series by plotting their sequences of partial sum. (14) Cauchy’s root test by plotting n -th roots. (15) Ratio test by plotting the ratio of n -th and $(n + 1)$ -th term.	
Suggested Readings		
Thiscoursecanbeoptedasanelectivebythestudentsoffollowingsubjects: Engg.andTech.(UG),Chemistry/Biochemistry/LifeSciences(UG), Economics(UG/PG), Commerce(UG), BBA/BCA, B.Sc.(C.S.)		
Suggested Continuous Evaluation Methods: Max. Marks: 25		
SN	Assessment Type	Max. Marks
1	Class Tests	10
2	Online Quizzes/ Objective Tests	5
3	Presentation	5
4	Assignment	5
Course prerequisites: To study this course, a student must have subject Mathematics in class 12 th		
Suggested equivalent online courses:		
Further Suggestions:		

B.A./B.Sc. I (SEMESTER-II) PAPER-I Matrices and Differential Equations & Geometry

Programme: Certificate Class: B.A./B.Sc.	Year: First	Semester: Second
Subject: Mathematics		
Course Code: B030201T	Course Title: Matrices and Differential Equations & Geometry	
Course outcomes:		
CO1: The subjects of the course are designed in such a way that they focus on developing mathematical skills in algebra, calculus and analysis and give in depth knowledge of geometry, calculus, algebra and other theories.		
CO2: The student will be able to find the rank, eigen values of matrices and study the linear homogeneous and non-homogeneous equations. The course in differential equation intends to develop problem solving skills for solving various types of differential equation and geometrical meaning of differential equation.		
CO3: The subjects learn and visualize the fundamental ideas about coordinate geometry and learn to describe some of the surface by using analytical geometry.		
CO4: On successful completion of the course students have gained knowledge about regular geometrical figures and their properties. They have the foundation for higher course in Geometry.		
Credits: 6	Core Compulsory / Elective	
Max. Marks: 25+75	Min. Passing Marks:	
Total No. of Lectures-Tutorials-Practical (in hours per week): L-T-P: 6-0-0		
PART-A Matrices and Differential Equations		
Unit	Topics	No. of Lectures
I	Types of Matrices, Elementary operations on Matrices, Rank of a Matrix, Echelon form of a Matrix, Normal form of a Matrix, Inverse of a Matrix by elementary operations, System of linear homogeneous and non-homogeneous equations, Theorems on consistency of a system of linear equations.	12
II	Eigen values, Eigen vectors and characteristic equation of a matrix, Cayley-Hamilton theorem and its use in finding inverse of a matrix, Complex functions and separation into real and imaginary parts, Exponential and Logarithmic functions, Inverse trigonometric and hyperbolic functions.	11
III	Formation of differential equations, Geometrical meaning of a differential equation, Equation of first order and first degree, Equation in which the variables are separable, Homogeneous equations, Exact differential equations and equations reducible to the exact form, Linear equations.	11
IV	First order higher degree equations solvable for x,y,p , Clairaut’s equation and singular solutions, orthogonal trajectories, Linear differential equation of order greater than one with constant coefficients, Cauchy- Euler form.	11

PART-B Geometry		
Unit	Topics	No. of Lectures
V	General equation of second degree, System of conics, Tracing of conics.	8
VI	Confocal conics, Polar equation of conics and its properties.	9
VII	Three-Dimensional Coordinates, Projection and Direction Cosine, Plane (Cartesian and vector form), Straight line in three dimensions.	13
VIII	Sphere, Cone and Cylinder.	15
Suggested Readings (PART-A Matrices and Differential Equations): - Shanti Narayan, A Textbook of Matrices, S. Chand, 2010 - Fuzhen Zhang, Matrix Theory- Basic Results and Techniques, Springer,1999 - Stephen H. Friedberg, A.J Insel& L.E. Spence, Linear Algebra,Pearson - William E Boyce and Richard C Di Prima, Elementary Differential Equations and Boundary Value Problems, John Wiley and Sons, 2009 B. Rai, D.P. Choudhary& H. J. Freedman, A Course in Differential Equations,Narosa - D.A. Murray, Introductory Course in Differential Equations, OrientLongman - Suggested digitalplatform:NPTEL/SWAYAM/MOOCs Suggested Readings (Part-B Geometry): - Robert J.T Bell, Elementary Treatise on Coordinate Geometry of three dimensions, Macmillan IndiaLtd. - P.R. Vittal, Analytical Geometry 2d & 3D,Pearson. - S.L. Loney, The Elements of Coordinate Geometry, McMillan and Company,London. - Suggested digitalplatform:NPTEL/SWAYAM/MOOCs		
This course can be opted as an elective by the students of following subjects:Engg.andTech.(UG),Economics(UG/PG),Commerce(UG),BBA/BCA, B.Sc.(C.S.)		
Suggested Continuous Evaluation Methods: Max. Marks: 25		
SN	Assessment Type	Max. Marks
1	Class Tests	10
2	Online Quizzes/ Objective Tests	5
3	Presentation	5
4	Assignment	5
Course prerequisites: To study this course, a student must have subject Mathematics in class 12 th		
Suggested equivalent online courses:		
Further Suggestions:		

B.A. /B.Sc. II (MATHEMATICS)

Detailed Syllabus For

DIPLOMA

IN

MATHEMATICS

B.A./B.Sc.II (SEMESTER-III) PAPER-I Algebra & Mathematical Methods

Programme: Diploma	Year: Second	Semester: Third
Class: B.A./B.Sc.		
Subject: Mathematics		
Course Code: B030301T	Course Title: Algebra & Mathematical Methods	
Courseoutcomes:		
CO1: Group theory is one of the building blocks of modern algebra. Objective of this course is to introduce students to basic concepts of Group, Ring theory and their properties.		
CO2: A student learning this course gets a concept of Group, Ring, Integral Domain and their properties. This course will lead the student to basic course in advanced mathematics and Algebra.		
CO3: The course gives emphasis to enhance students’ knowledge of functions of two variables, Laplace Transforms, Fourier Series.		
CO4: On successful completion of the course students should have knowledge about higher different mathematical methods and will help him in going for higher studies and research.		
Credits: 6	Core Compulsory / Elective	
Max. Marks: 25+75	Min. Passing Marks:	
Total No. of Lectures-Tutorials-Practical (in hours per week): L-T-P: 6-0-0		
Part- A		
Algebra		
Unit	Topics	No. of Lectures
I	Introduction to Indian ancient Mathematics and Mathematicians should be included under Continuous Internal Evaluation (CIE). Equivalence relations and partitions, Congruence modulo n, Definition of a group with examples and simple properties, Subgroups, Generators of a group, Cyclic groups.	12
II	Permutation groups, Even and odd permutations, The alternating group, Cayley’s theorem, Direct products, Coset decomposition, Lagrange’s theorem and its consequences, Fermat and Euler theorems	11
III	Normal subgroups, Quotient groups, Homomorphism and isomorphism, Fundamental theorem of homomorphism, Theorems on isomorphism.	11
IV	Rings, Subrings, Integral domains and fields, Characteristic of a ring, Ideal and quotient rings, Ring homomorphism, Field of quotient of an integral domain.	11

Part- B Mathematical Methods		
Unit	Topics	No. of Lectures
V	Limit and Continuity of functions of two variables, Differentiation of function of two variables, Necessary and sufficient condition for differentiability of functions two variables, Schwarz’s and Young theorem, Taylor's theorem for functions of two variables with examples, Maxima and minima for functions of two variables, Lagrange multiplier method, Jacobians.	12
VI	Existence theorems for Laplace transforms, Linearity of Laplace transform and their properties, Laplace transform of the derivatives and integrals of a function, Convolution theorem, inverse Laplace transforms, Solution of the differential equations using Laplace transforms.	11
VII	Fourier series, Fourier expansion of piecewise monotonic functions, Half and full range expansions, Fourier transforms (finite and infinite), Fourier integral.	11
VIII	Calculus of variations-Variational problems with fixed boundaries- Euler's equation for functionals containing first order derivative and one independent variable, Extremals, Functionals dependent on higher order derivatives, Functionals dependent on more than one independent variable, Variational problems in parametric form.	11
Suggested Readings(Part-A Algebra): 1. J.B. Fraleigh, A first course in Abstract Algebra, Addison-weley 2. I. N. Herstein, Topics in Algebra, John Wiley & Sons 3. Joseph A Gallian, Contemporary Abstract Algebra, Brooks/Cole Cengage Learning, 2016 4. V K Khanna and S K Bhambri, A course in Abstract Algebra, Vikas Publishing House, 2016 5. Suggested digital platform:NPTEL/SWAYAM/MOOCs Suggested Readings (Part- B Mathematical Methods): 1. T.M. Apostol, Calculus Vol. II, John Wiley and Sons Inc. 2. G.F. Simmons, Differential Equations with Application and Historical Notes, Tata-McGraw Hill 3. M. D. Raisinghania, Advanced Differential Equations, S. Chand Publication. 4. Erwin Kreyszig, Advanced Engineering Mathematics, John Wiley & Sons. 5. Suggested digital platform: NPTEL/SWAYAM/MOOCs 6. Course Books published in Hindi may be prescribed by the Universities.		
This course can be opted as an elective by the students of following subjects: Engg. and Tech. (UG), B.Sc.(C.S.)		
Suggested Continuous Evaluation Methods: Max. Marks: 25		
SN	Assessment Type	Max. Marks
1	Class Tests	10
2	Online Quizzes/ Objective Tests	5
3	Presentation	5
4	Assignment (Introduction to Indian ancient Mathematics and Mathematicians)	5
Course prerequisites: To study this course, a student must have subject Mathematics in class 12 th		
Suggested equivalent online courses:		
Further Suggestions:		

B.A./B.Sc. II (SEMESTER-IV) PAPER-I Differential Equations & Mechanics

Programme: Diploma Class: B.A./B.Sc.	Year: Second	Semester: Fourth
Subject: Mathematics		
Course Code: B030401T	Course Title: Differential Equations & Mechanics	
Course outcomes: CO1: The objective of this course is to familiarize the students with various methods of solving differential equations, partial differential equations of first order and second order and to have qualitative applications. CO2: A student doing this course is able to solve differential equations and is able to model problems in nature using ordinary differential equations. After completing this course, a student will be able to take more courses on wave equation, heat equation, diffusion equation, gas dynamics, non linear evolution equation etc. These entire courses are important in engineering and industrial applications for solving boundary value problem. CO3: The object of the paper is to give students knowledge of basic mechanics such as simple harmonic motion, motion under other laws and forces. CO4: The student, after completing the course can go for higher problems in mechanic such as hydrodynamics, this will be helpful in getting employment in industry.		
Credits: 6	Core Compulsory / Elective	
Max. Marks: 25+75	Min. Passing Marks:	
Total No. of Lectures-Tutorials-Practical (in hours per week): L-T-P: 6-0-0		
Part- A Differential Equations		
Unit	Topics	No. of Lectures
I	Second order linear differential equations with variable coefficients: Use of a known solution to find another, normal form, method of undetermined coefficient, variation of parameters, Series solutions of differential equations, Power series method.	12
II	Bessel, Legendre and Hypergeometric functions and their properties, recurrence and generating relations.	10
III	Origin of first order partial differential equations. Partial differential equations of the first order and degree one, Quasi-linear PDE and its solution using Lagrange's method, Partial differential equation of first order and degree greater than one. Charpit's method of solution, Surfaces Orthogonal to the given system of surfaces.	12
IV	Origin of second order PDE, Solution of partial differential equations of the second and higher order with constant coefficients, Classification of linear partial differential equations of second order, Solution of second order partial differential equations with variable coefficients.	11

Part- B Mechanics		
Unit	Topics	No. of Lectures
V	Analytical conditions of equilibrium of Coplanar forces, Forces in three dimensions, Poinot's central axis, Wrenches, Null lines and planes.	12
VI	Virtual work, Stable and Unstable equilibrium, Common Catenary.	11
VII	Velocities and accelerations along radial and transverse directions, and along tangential and normal directions, Simple Harmonic motion, Motion under other law of forces. Elastic strings, Motion in resisting medium.	11
VIII	Constrained motion, Motion on smooth and rough plane curves, Motion of particles of varying mass, Rocket motion, Central orbit, Kepler's laws of motion, Motion of particle in three dimensions.	11
Suggested Readings(Part-A Differential Equations): 1. G.F. Simmons, Differential Equations with Application and Historical Notes, Tata–McGraw Hill. 2. M. D. Raisinghania, Introduction to ordinary and Partial Differential equations, S. Chand Publication. 3. B. Rai, D.P. Choudhary & H. J. Freedman, A Course of Ordinary Differential Equations, Narosa 4. Ian N. Snedden, Elements of Partial Differential Equations, Dover Publication. 5. L.E. Elsgolts, Differential Equation and Calculus of variations, University Press of thePacific. 6. Suggested digitalplatform:NPTEL/SWAYAM/MOOCs Suggested Readings(Part-B Mechanics): 1. R.C. Hibbeler, Engineering Mechanics-Statics, Prentice Hall Publishers 2. R.C. Hibbeler, Engineering Mechanics-Dynamics, Prentice Hall Publishers 3. A. Nelson, Engineering Mechanics Statics and Dynamics, Tata McGraw Hill 4. J.L. Synge & B.A. Griffith, Principles of Mechanics, Tata McGraw Hill 5. M Ray, A Textbook on Dynamics, S Chand. 6. R S Verma, A Text Book of Statics, Pothishala. 7. Suggested digitalplatform:NPTEL/SWAYAM/MOOCs		
This course can be opted as an elective by the students of following subjects: Engg. and Tech. (UG), Economics(UG/PG),B.Sc.(C.S.)		
Suggested Continuous Evaluation Methods: Max. Marks: 25		
SN	Assessment Type	Max. Marks
1	Class Tests	10
2	Online Quizzes/ Objective Tests	5
3	Presentation	5
4	Assignment	5
Course prerequisites: To study this course, a student must have Certificate Course in Applied Mathematics		
Suggested equivalent online courses:		
Further Suggestions:		

B.A. /B.Sc. III (MATHEMATICS)

Detailed Syllabus For

**DEGREE IN
MATHEMATICS**

B.A./B.Sc. III (SEMESTER-V) PAPER-I Group and Ring Theory & Linear Algebra

Programme: Degree		Year: Third	Semester: Fifth	
Class: B.A./B.Sc.				
Subject: Mathematics				
Course Code: B030501T		Course Title: Group and Ring Theory & Linear Algebra		
Course outcomes:				
CO1: Liner algebra is a basic course in almost all branches of science. The objective of this course is to introduce a student to the basics of linear algebra and some of its applications.				
CO2: Students will be able to know the concepts of group, ring and other related properties which will prepare the students to take up further applications in the relevant fields.				
CO3: The student will use this knowledge in computer science, finance mathematics, industrial mathematics and bio mathematics. After completion of this course students appreciate its interdisciplinary nature.				
Credits: 5		Core Compulsory / Elective		
Max. Marks: 25+75		Min. Passing Marks:		
Total No. of Lectures-Tutorials-Practical (in hours per week): L-T-P: 5-0-0				
PART-A				
Group and Ring Theory				
Unit	Topics			No. of Lectures
I	Introduction to Indian ancient Mathematics and Mathematicians should be included under Continuous Internal Evaluation (CIE). Automorphism, inner automorphism, Automorphism groups, Automorphism groups of finite and infinite cyclic groups, Characteristic subgroups, Commutator subgroup and its properties; Applications of factor groups to automorphism groups.			10
II	Conjugacy classes, The class equation, p -groups, The Sylow theorems and consequences, Applications of Sylow theorems; Finite simple groups, Nonsimplicity tests; Generalized Cayley’s theorem, Index theorem, Embedding theorem and applications.			10
III	Polynomial rings over commutative rings, Division algorithm and consequences, Principal ideal domains, Factorization of polynomials, Reducibility tests, Irreducibility tests, Eisenstein criterion, Unique factorization in $\mathbb{Z}[x]$.			9
IV	Divisibility in integral domains, Irreducibles, Primes, Unique factorization domains, Euclidean domains.			9

PART-B Linear Algebra		
Unit	Topics	No. of Lectures
V	Vector spaces, Subspaces, Linear independence and dependence of vectors, Basis and Dimension, Quotient space.	10
VI	Linear transformations, The Algebra of linear transformations, rank nullity theorem, their representation as matrices.	9
VII	Linear functionals, Dual space, Characteristic values, Cayley Hamilton Theorem.	9
VIII	Inner product spaces and norms, Cauchy-Schwarz inequality, Orthogonal vectors, Orthonormal sets and bases, Bessel’s inequality for finite dimensional spaces, Gram-Schmidt orthogonalization process, Bilinear and Quadratic forms.	9
Suggested Readings: 1. Topics in Algebra by I. N.Herstein. 2. Contemporary Abstract Algebra by Joseph A Gallian. 3. Linear Algebra by K. Hoffman and R.Kunze. 4. Linear Algebra by S H Friedberg, A J Insel and L E Spence. 5. Suggested digital platform: NPTEL/SWAYAM/MOOCs		
This course can be opted as an elective by the students of following subjects: Engg. and Tech. (UG), BCA, B.Sc.(C.S.)		
Suggested Continuous Evaluation Methods: Max. Marks: 25		
SN	Assessment Type	Max. Marks
1	Class Tests	10
2	Online Quizzes/ Objective Tests	5
3	Presentation	5
4	Assignment (Introduction to Indian ancient Mathematics and Mathematicians)	5
Course prerequisites: To study this course, a student must have Diploma in Mathematics		
Suggested equivalent online courses:		
Further Suggestions:		

B.A./B.Sc. III (SEMESTER-V) PAPER-II (i) Number Theory & Game Theory

Programme: Degree		Year: Third	Semester: Sixth	
Class: B.A./B.Sc.				
Subject: Mathematics				
Course Code: B030502T		Course Title: Number Theory &Game Theory		
Course outcomes:				
CO1: Upon successful completion, students will have the knowledge and skills to solve problems in elementary number theory and also apply elementary number theory to cryptography.				
CO2: This course provides an introduction to Game Theory. Game Theory is a mathematical framework which makes possible the analysis of the decision making process of interdependent subjects. It is aimed at explaining and predicting how individuals behave in a specific strategic situation, and therefore help improve decision making.				
CO3: A situation is strategic if the outcome of a decision problem depends on the choices of more than one person. Most decision problems in real life are strategic.				
CO4: To illustrate the concepts, real-world examples, case studies, and classroom experiments might be used.				
Credits: 5		Core Compulsory / Elective		
Max. Marks: 25+75		Min. Passing Marks:		
Total No. of Lectures-Tutorials-Practical (in hours per week): L-T-P: 5-0-0				
Part- A				
Number Theory				
Unit	Topics			No. of Lectures
I	Theory of Numbers Divisibility; Euclidean algorithm; primes; congruences; Fermat’s theorem, Euler’s theorem and Wilson’s theorem; Fermat’s quotients and their elementary consequences; solutions of congruences; Chinese remainder theorem; Euler’sphi-function.			10
II	Congruences Congruence modulo powers of prime; primitive roots and their existence; quadratic residues; Legendre symbol, Gauss’ lemma about Legendre symbol; quadratic reciprocity law; proofs of various formulations; Jacobisymbol.			9
III	Diophantine Equations Solutions of $ax + by = c, x^n + y^n = z^n$; properties of Pythagorean triples; sums of two, four and five squares; assorted examples of diophantine equations.			9
IV	Generating Functions and Recurrence Relations Generating Function Models, Calculating coefficient of generating functions, Partitions, Exponential Generating Functions, A Summation Method. Recurrence Relations: Recurrence Relation Models, Divide and conquer Relations, Solution of Linear, Recurrence Relations, Solution of Inhomogeneous Recurrence Relations, Solutions with GeneratingFunctions.			9

Part- B Game Theory		
Unit	Topics	No. of Lectures
V	Introduction, overview, uses of game theory, some applications and examples, and formal definitions of: the normal form, payoffs, strategies, pure strategy Nash equilibrium.	10
VI	Introduction, characteristic of game theory, Two- person zero-sum game, Pure and Mixed strategies, Saddle point and its existence.	10
VII	Fundamental Theorem of Rectangular games, Concept of Dominance, Dominance and Graphical method of solving Rectangular games.	9
VIII	Relationship between rectangular game and Linear Programming Problem, Solving rectangular game by Simplex method, reduction of $m \times n$ game and solution of 2×2 , $2 \times s$ and $r \times 2$ cases by graphical method, algebraic and linear programming solution of $m \times n$ games.	9
Suggested Readings (Part-A Number Theory): - Niven, I., Zuckerman, H. S. and Montgomery, H. L. (2003) An Int. to the Theory of Numbers (6th edition) John Wiley and sons, Inc., NewYork. - Burton, D. M. (2002) Elementary Number Theory (4th edition) Universal Book Stall, NewDelhi. - Balakrishnan, V. K. (1994) Schaum’s Outline of Theory and Problems of Combinatorics Including Concepts of Graph Theory, Schaum’sOutline. - Balakrishnan,V. K. (1996) Introductory Discrete Mathematics, DoverPublications. - Suggested digitalplatform:NPTEL/SWAYAM/MOOCs Suggested Readings (Part-B Game Theory): - Martin Osborne, An Introduction to Game Theory, Oxford University Press,2003 - Vijay Krishna, Game Theory, AcademicPress. - Prajit Dutta, Strategies and Games, MIT Press, (Website 1)http://www.ece.stevens-tech.edu/~ccomanic/ee800c.html - Allan MacKenzie, Game Theory for Wireless Engineers, Synthesis lectures on Communications,2006 - Suggested digitalplatform:NPTEL/SWAYAM/MOOCs		
This course can be opted as an elective by the students of following subjects: Engg. and Tech. (UG), B.Sc.(C.S.)		
Suggested Continuous Evaluation Methods: Max. Marks: 25		
SN	Assessment Type	Max. Marks
1	Class Tests	10
2	Online Quizzes/ Objective Tests	5
3	Presentation	5
4	Assignment	5
Course prerequisites: To study this course, a student must have Diploma in Mathematics		
Suggested equivalent online courses:		
Further Suggestions:		

B.A./B.Sc. III (SEMESTER-V) PAPER-II (ii) Graph Theory & Discrete Mathematics

Programme: Degree Class: B.A./B.Sc.	Year: Third	Semester: Sixth
Subject: Mathematics		
Course Code: B030502T	Course Title: Graph Theory & Discrete Mathematics	
Course outcomes:		
CO1: Upon successful completion, students will have the knowledge of various types of graphs, their terminology and applications.		
CO2: After Successful completion of this course students will be able to understand the isomorphism and homomorphism of graphs. This course covers the basic concepts of graphs used in computer science and other disciplines. The topics include path, circuits, adjacency matrix, tree, coloring.. After successful completion of this course the student will have the knowledge graph coloring, color problem, vertex coloring.		
CO3: After successful completion, students will have the knowledge of Logic gates, Karnaugh maps and skills to proof by using truth tables. After Successful completion of this course students will be able to apply the basics of the automation theory, transition function and table.		
CO4: This course covers the basic concepts of discrete mathematics used in computer science and other disciplines that involve formal reasoning. The topics include logic, counting, relations, hasse diagram and Boolean algebra. After successful completion of this course the student will have the knowledgein Mathematical reasoning, combinatorial analysis, discrete structures and Applications.		
Credits: 5	Core Compulsory / Elective	
Max. Marks: 25+75	Min. Passing Marks:	
Total No. of Lectures-Tutorials-Practical (in hours per week): L-T-P: 5-0-0		
Part- A Graph Theory		
Unit	Topics	No. of Lectures
I	Introduction to graphs, basic properties of graphs, Simple graph, multi graph, graph terminology, representation of graphs, Bipartite, regular, planar and connected graphs, connected components in a graph, Euler graphs, Directed, Undirected, multi-graph, mixed graph.	10
II	Walk and unilateral components, unicursal graph, Hamiltonian path and circuits, Graph colouring, chromatics number, isomorphism and homomorphism of graphs, Incidence relation and degree of the graph.	9
III	Operation of graph circuit, Path and circuits, Eulerian circuits, Hamiltonian path and cycles, Adjacency matrix, Weighted graph, Travelling salesman problem, Shortest path, Dijkstra’s algorithm.	9
IV	Tree, Binary and Spanning trees, Coloring, Color problems, Vertex coloring and important properties.	9

Part- B Discrete Mathematics		
Unit	Topics	No. of Lectures
V	Propositional Logic- Proposition logic, basic logic, logical connectives, truth tables, tautologies, contradiction, normal forms (conjunctive and disjunctive), modus ponens and modus tollens, validity, predicate logic, universal and existential quantification, proof by implication, converse, inverse contrapositive, contradiction, direct proof by using truth table. Relation- Definition, types of relation, domain and range of a relation, pictorial representation of relation, properties of relation, partial ordering relation.	10
VI	Boolean Algebra- Basic definitions, Sum of products and products of sums, Logic gates and Karnaugh maps. Graphs- Simple graph, multi graph, graph terminology, representation of graphs, Bipartite, regular, planar and connected graphs, connected components in a graph, Euler graphs, Hamiltonian path and circuits, Graph colouring, chromatics number, isomorphism and homomorphism of graphs.	10
VII	Combinatories- Inclusion- exclusion, recurrence relations (nth order recurrence relation with constant coefficients, Homogeneous recurrence relations, Inhomogeneous recurrence relations), generating function (closed form expression, properties of G.F., solution of recurrence relations using G.F. solution of combinatorial problem using G.F.)	9
VIII	Finite Automata- Basic concepts of automation theory, Deterninistic Finite Automation (DFA), transition function, transitionable, Non Deterministic Finite Automata (NDFA), Mealy and Moore machine, Minimization of finite automation.	9
Suggested Readings (Part-A Graph Theory): 1. “Graph Theory with Applications to Engineering and Computer Science” by NarsinghDeo 2. “Introduction to Graph Theory” by Douglas B West 3. “Graph Theory with Algorithms and Its Applications: In Applied Science and Technology” by SantanuSaha Ray 4. Suggested digitalplatform:NPTEL/SWAYAM/MOOCs. Suggested Readings (Part-B Discrete Mathematics): 1. Discrete Mathematics by C.L.Liu. 2. Discrete Mathematics with computer application by Trembley and Manohar. 3. Discrete Mathematics and Its Applications by Kenneth H.Rosen 4. Suggested digital platform: NPTEL/SWAYAM/MOOCs		
This course can be opted as an elective by the students of following subjects: Engg. and Tech. (UG), B.Sc.(C.S.)		
Suggested Continuous Evaluation Methods: Max. Marks: 25		
SN	Assessment Type	Max. Marks
1	Class Tests	10
2	Online Quizzes/ Objective Tests	5
3	Presentation	5
4	Assignment	5
Course prerequisites: To study this course, a student must have Diploma in Mathematics		
Suggested equivalent online courses:		
Further Suggestions:		

B.A./B.Sc. III (SEMESTER-V) PAPER-II (iii) Differential Geometry & Tensor Analysis

Programme: Degree Class: B.A./B.Sc.		Year: Third	Semester: Sixth	
Subject: Mathematics				
Course Code: B030502T		Course Title: Differential Geometry & Tensor Analysis		
Course outcomes:				
CO1: After Successful completion of this course, students should be able to determine and calculate curvature of curves in different coordinate systems.				
CO2: This course covers the Local theory of Curves, Local theory of surfaces, Geodesics, Geodesics curvature, Geodesic polars, Curvature of curves on surfaces, Gaussian curvature, Normal curvature etc.				
CO3: After Successful completion of this course, students should have the knowledge of tensor algebra, different types of tensors, Riemannian space, Ricci tensor, Einstein space and Einstein tensor etc.				
Credits: 5		Core Compulsory / Elective		
Max. Marks: 25+75		Min. Passing Marks:		
Total No. of Lectures-Tutorials-Practical (in hours per week): L-T-P: 5-0-0				
Part- A				
Differential Geometry				
Unit	Topics			No. of Lectures
I	Local theory of curves- Space curves, Examples, Plane Curves, tangent and normal and binormal, Osculating Plane, normal plane and rectifying plane, Osculating circle, osculating sphere Helices, Serret-Frenet apparatus, contact between curve and surfaces, tangent surfaces, involutes and evolutes of curves, Bertrand curves, Intrinsic equations, fundamental existence theorem for space curves.			10
II	Local Theory of Surfaces- Parametric patches on surface curve of a surface, family of surfaces (one parameter), edge of regression, ruled surfaces, skew ruled surfaces and developable surfaces, surfaces of revolution, Helicoids.			9
III	Metric-first fundamental form and arc length, Direction coefficients, families of curves, intrinsic properties, geodesics, canonical geodesic equations, normal properties of geodesics, geodesics curvature, Geodesic polars.			9
IV	Curvature of curves on surfaces, Gaussian curvature, normal curvature, Meusnier’s theorem, mean curvature, umbilic points, lines of curvature, Rodrigue’s formula, Euler’s theorem, Gauss-Bonnet theorem.			9

Part- B Tensor Analysis		
Unit	Topics	No. of Lectures
V	Tensor algebra: Vector spaces, the dual spaces, tensor product of vector spaces, transformation formulae, contraction, special tensors-symmetric tensor, inner product, associated tensor with examples.	10
VI	Tensor Analysis: Contravariant and covariant vectors and tensors, Mixed tensors, Symmetric and skew-symmetric tensors, Algebra of tensors, Contraction and inner product, Quotient theorem, Reciprocal tensors, Christoffel’s symbols, Law of transformation of Christoffel’s symbols, Covariant differentiation, non- commutativity of Covariant derivative.	10
VII	Gradient of scalars, Divergence of a contravariant vector, covariant vector and conservative vectors, Laplacian of an invariant, curl of a covariant vector, irrotational vector, with examples.	9
VIII	Riemannian space, Riemannian curvatures and their properties, geodesics, geodesic curvature, geometrical interpretation of curvature tensor, Ricci tensor, scalar curvature, Einstein space and Einstein tensor.	9
Suggested Readings (Part-A Differential Geometry): 1. T.J. Willmore, An Introduction to Differential Geometry, Dover Publications,2012. 2. B. O’Neill, Elementary Differential Geometry, 2nd Ed., Academic Press,2006. 3. C.E. Weatherburn, Differential Geometry of Three Dimensions, Cambridge University Press2003. 4. D.J. Struik, Lectures on Classical Differential Geometry, Dover Publications,1988. 5. S. Lang, Fundamentals of Differential Geometry, Springer,1999. 6. B. Spain, Tensor Calculus: A Concise Course, Dover Publications,2003. 7. L. P. Eisenhart, An Introduction to Differential Geometry (with the use of tensor Calculus), Princeton University Press,1940. 8. Tensor Analysis, Theory and Applications to Geometry and Mechanics of Continua, 2nd Edition, I. S. Sokolnikoff, John Wiley and Sons.,1964. 9. Suggested digital platform: NPTEL/SWAYAM/MOOCs Suggested Readings (Part-B Tensor Analysis): 1. Tensors- Mathematics of Differential Geometry by Z. Ahsan,PHI,2015 2. David C. Kay, Tensor Analysis, Schaum's Outline Series, McGraw Hill1988. 3. R. S, Mishra, A Course in Tensors with Applications to Reimannian Geometry, Pothishala Pvt. Ltd,Allahabad. 4. Suggested digital platform: NPTEL/SWAYAM/MOOCs		
This course can be opted as an elective by the students of following subjects: Engg. and Tech. (UG), B.Sc.(C.S.)		
Suggested Continuous Evaluation Methods: Max. Marks: 25		
SN	Assessment Type	Max. Marks
1	Class Tests	10
2	Online Quizzes/ Objective Tests	5
3	Presentation	5
4	Assignment	5
Course prerequisites: To study this course, a student must have Diploma in Mathematics		
Suggested equivalent online courses:		
Further Suggestions:		

B.A./B.Sc. III (SEMESTER-V) PAPER-III Research Project/Dissertation

Programme: Degree Class: B.A./B.Sc.	Year: Third	Semester: Fifth
Subject: Mathematics		
Course Code: B030503R	Course Title: Research Project /Dissertation	
Max Marks: 100	Core: Compulsory	
Course Outcomes(COs)		
1. Develop the ability to conduct independent research in specific area of mathematics. 2. Enhance critical thinking skills by analysing and evaluating existing literature and experimental data. 3. Demonstrate the capacity to solve complex mathematical problems related to the chosen topic. 4. Improve scientific writing skills, including the ability to clearly and concisely communicate research findings. 5. Gain proficiency in data analysis techniques and the use of relevant software and techniques. 6. Present research findings through oral presentations and written reports. 7. Make a unique contribution to the field by adding new insights, findings, or experimental data. 8. Understand and adhere to ethical standards in research, including proper citation and plagiarism avoidance.		

B.A./B.Sc. III (SEMESTER-VI) PAPER-I METRIC SPACES & COMPLEX ANALYSIS

Programme: Degree		Year: Third	Semester: Sixth
Class: B.A./B.Sc.			
Subject: Mathematics			
Course Code: B030601T		Course Title: METRIC SPACES & COMPLEX ANALYSIS	
Course outcomes:			
CO1: The course is aimed at exposing the students to foundations of analysis which will be useful in understanding various physical phenomena and gives the student the foundation in mathematics.			
CO2: After completion of this course the student will have rigorous and deeper understanding of fundamental concepts in Mathematics. This will be helpful to the student in understanding pure mathematics and in research.			
CO3: Students will be able to know the concepts of metric space, basic concepts and developments of complex analysis which will prepare the students to take up further applications in the relevant fields.			
Credits: 4		Core Compulsory / Elective	
Max. Marks: 25+75		Min. Passing Marks:	
Total No. of Lectures-Tutorials-Practical (in hours per week): L-T-P: 4-0-0			
Part- A			
Metric Spaces			
Unit	Topics		No. of Lectures
I	Basic Concepts Metric spaces: Definition and examples, Sequences in metric spaces, Cauchy sequences, Complete metric space.		8
II	Topology of Metric Spaces Open and closed ball, Neighborhood, Open set, Interior of a set, limit point of a set, derived set, closed set, closure of a set, diameter of a set, Cantor’s theorem, Subspaces, Dense set, Countability, Nowhere dense sets.		8
III	Continuity & Uniform Continuity in Metric Spaces Continuous mappings, Sequential criterion and other characterizations of continuity, Uniform continuity, Homeomorphism, Isometry, Equivalent metrics, Contraction mapping, Banach fixed point theorem.		7
IV	Connectedness and Compactness Connectedness, Connected subsets of $\mathbb{R}$, Connectedness and continuous mappings, Compactness, Compactness and boundedness, Continuous functions on compact spaces.		7

Part- B Complex Analysis		
Unit	Topics	No. of Lectures
V	Analytic Functions and Cauchy-Riemann Equations Functions of complex variable, Mappings; Mappings by the exponential function, Limits, Theorems on limits, Limits involving the point at infinity, Continuity, Complex differentiability, Cauchy-Riemann equations, Sufficient conditions for differentiability; Analytic functions and their examples.	8
VI	Elementary Functions and Integrals Exponential function, Logarithmic function, Trigonometric function, Derivatives of functions, Definite integrals of functions, Contours, Contour integrals and its examples, Upper bounds for moduli of contour integrals.	8
VII	Cauchy’s Theorems and Fundamental Theorem of Algebra Anti-derivatives, Proof of anti derivative theorem, Cauchy-Goursat theorem, Cauchy integral formula; An extension of Cauchy integral formula, Consequences of Cauchy integral formula, Liouville’s theorem and the fundamental theorem of algebra.	7
VIII	Series and Residues Convergence of sequences and series, Taylor series and its examples; Laurent series and its examples, Absolute and uniform convergence of power series, Uniqueness of series representations of power series, Isolated singular points, Residues, Cauchy’s residue theorem, residue at infinity; Types of isolated singular points, Residues at poles and its examples.	7
Suggested Readings (Part-A Metric Space): 1. Mathematical Analysis by Shanti Narain. 2. Shirali, Satish & Vasudeva, H. L. (2009). Metric Spaces , Springer, First Indian Print. 3. Kumaresan, S. (2014). Topology of Metric Spaces (2nd ed.). Narosa Publishing House. New Delhi. 4. Simmons, G. F. (2004). Introduction to Topology and Modern Analysis . Tata McGraw Hill. New Delhi. 5. Suggested digital platform: NPTEL/SWAYAM/MOOCs.		
Suggested Readings (Part-B Complex Analysis): 1. Function of Complex Variable by Shanti Narain. 2. Complex variable and applications by Brown & Churchill. 3. Functions of Complex Analysis by S Ponnusamy. 4. Suggested digital platform: NPTEL/SWAYAM/MOOCs.		
This course can be opted as an elective by the students of following subjects: Engg. and Tech. (UG), B.Sc.(C.S.)		
Suggested Continuous Evaluation Methods: Max. Marks: 25		
SN	Assessment Type	Max. Marks
1	Class Tests	10
2	Online Quizzes/ Objective Tests	5
3	Presentation	5
4	Assignment	5
Course prerequisites: To study this course, a student must have Diploma in Mathematics		
Suggested equivalent online courses:		
Further Suggestions:		

B.A./B.Sc. III (SEMESTER-VI) PAPER-II Numerical Analysis & Operation Research

Programme: Degree		Year: Third	Semester: Sixth	
Class: B.A./B.Sc.				
Subject: Mathematics				
Course Code: B030602T		Course Title: Numerical Analysis & Operations Research		
Course outcomes:				
CO1: The aim of this course is to teach the student the application of various numerical technique for variety of problems occurring in daily life. At the end of the course the student will be able to understand the basic concept of Numerical Analysis and to solve algebraic and differential equation.				
CO2: The main outcome will be that students will be able to handle problems and finding approximated solution. Later he can opt for advance course in Numerical Analysis in higher Mathematics.				
CO3: The student will be able to solve various problems based on convex sets and linear programming. After successful completion of this paper will enable the students to apply the basic concepts of transportation problems and its related problems to apply in further concepts and application of operations research.				
Credits: 4		Core Compulsory / Elective		
Max. Marks: 25+75		Min. Passing Marks:		
Total No. of Lectures-Tutorials-Practical (in hours per week): L-T-P: 4-0-0				
PART-A				
Numerical Analysis				
Unit	Topics			No. of Lectures
I	Solution of equations: bisection, Secant, Regular Falsi, Newton Raphson’s method, Newton’s method for multiple roots, Interpolation, Lagrange and Hermite interpolation, Difference schemes, Divided differences, Interpolation formula using differences.			8
II	Numerical differentiation, Numerical Quadrature: Newton Cotes Formulas, Gaussian Quadrature Formulas, System of Linear equations: Direct method for solving systems of linear equations (Gauss elimination, LU Decomposition, Cholesky Decomposition), Iterative methods (Jacobi, Gauss Seidel, Relaxation methods). The Algebraic Eigen value problem: Jacobi’s method, Givens method, Power method.			8
III	Numerical solution of Ordinary differential equations: Euler method, single step methods, Runge-Kutta method, Multi-step methods: Milne-Simpson method, Types of approximation: Last Square polynomial approximation, Uniform approximation, Chebyshev polynomial approximation.			7
IV	Difference Equations and their solutions, Shooting method and Difference equation method for solving Linear second order differential equation with boundary conditions of first, second and third type.			7

PART-B Operations Research		
Unit	Topics	No. of Lectures
V	Introduction, Linear programming problems, statement and formation of general linear programming problems, graphical method, slack and surplus variables, standard and matrix forms of linear programming problem, basic feasible solution.	8
VI	Convex sets, fundamental theorem of linear programming, basic solution, Simplex method, introduction to artificial variables, two phase method, Big-M method and their comparison.	8
VII	Resolution of degeneracy, duality in linear programming problems, primal dual relationships, revised simplex method, sensitivity analysis.	7
VIII	Transportation problems, assignment problems.	7
Suggested Readings(Part-A Numerical Analysis): 1. Numerical Methods for Engineering and scientific computation by M. K. Jain, S.R.K. Iyengar& R.K.Jain. 2. Introductory methods of Numerical Analysis by S. S.Sastry 3. Elementary Numerical Analysis An Algorithmic Approach by Samuel D. Conte and Carl de Boor 4. Suggested digitalplatform:NPTEL/SWAYAM/MOOCs Suggested Readings(Part-B Operation Research): 1.Taha, Hamdy H, " Opearations Research- An Introduction ", PearsonEducation. 2.KantiSwarup , P. K. Gupta , Man Mohan, Operations research , Sultan Chand &Sons 3.Hillier Frederick S and Lieberman Gerald J., “ Operations Research ”, McGraw HillPublication. 4. Winston Wayne L., “ Operations Research: Applications and Algorithms ”, Cengage Learning, 4 th Edition. 5.Hira D.S. and Gupta Prem Kumar, “ Problems in Operations Research: Principles and Solutions ”, S Chand & CoLtd. 6. Kalavathy S., “ Operations Research ”, SChand. 7. Suggested digitalplatform:NPTEL/SWAYAM/MOOCs. This course can be opted as an elective by the students of following subjects: Engg. and Tech. (UG), Economics(UG/PG), B.Sc.(C.S.)		
Suggested Continuous Evaluation Methods: Max. Marks: 25		
SN	Assessment Type	Max. Marks
1	Class Tests	10
2	Online Quizzes/ Objective Tests	5
3	Presentation	5
4	Assignment	5
Course prerequisites: To study this course, a student must have Certificate Course in Applied Mathematics		
Suggested equivalent online courses:		
Further Suggestions:		

B.A./B.Sc. III (SEMESTER-VI) PAPER-III Practical

Programme: Degree		Year: Third	Semester: Sixth	
Class: B.A./B.Sc.				
Subject: Mathematics				
Course Code: B030603P		Course Title: Practical		
Course outcomes:				
The main objective of the course is to equip the student to solve the transcendental and algebraic equations, system of linear equations, ordinary differential equations,Interpolation,NumericalIntegration,MethodoffindingEigenvaluebyPowermethod(upto4×4),FittingaPolynomialFunction(uptothird degree).				
Credits: 2		Core Compulsory / Elective		
Max. Marks: 25+75		Min. Passing Marks:		
Total No. of Lectures-Tutorials-Practical (in hours per week): L-T-P: 0-0-4				
Unit	Topics			No. of Lectures
	Practical / Lab work to be performed in Computer Lab. List of the practicals to be done using computer algebra software (CAS), for example Mathematica/MATLAB/Maple/Maxima/Scilab/Octave etc 1. Solution of transcendental and algebraic equations by i) Bisection method ii) Newton Raphson method (Simple root, multiple roots, complexroots). iii) Secant method. iv) Regula Falsi method. 2. Solution of system of linear equations i) LU decomposition method ii) Gaussian elimination method iii) Gauss-Jacobi method iv) Gauss-Seidel method 3. Interpolation i) Lagrange Interpolation ii) Newton’s forward, backward and divided difference interpolations 4. Numerical Integration i) Trapezoidal Rule ii) Simpson’s one third rule iii) Weddle’s Rule iv) Gauss Quadrature 5. Method of finding Eigenvalue by Power method (up to 4 × 4) 6. Fitting a Polynomial Function (up to third degree)			

	7. Solution of ordinary differential equations i) Euler method ii) Modified Euler method iii) Runge Kutta method (order4) (iv) The method of successive approximations (Picard)	
Suggested Readings:		
This course can be opted as an elective by the students of following subjects: Engg. and Tech. (UG), Economics(UG/PG), B.Sc.(C.S.)		
Suggested Continuous Evaluation Methods: Max. Marks: 25		
SN	Assessment Type	Max. Marks
1	Class Tests	10
2	Online Quizzes/ Objective Tests	5
3	Presentation	5
4	Assignment	5
Course prerequisites: To study this course, a student must have Certificate Course in Applied Mathematics		
Suggested equivalent online courses:		
Further Suggestions:		

B.A./B.Sc. III (SEMESTER-VI) PAPER-IV Research Project/Dissertation

Programme: Degree Class: B.A./B.Sc.	Year: Third	Semester: Sixth
Subject: Mathematics		
Course Code: B030604R	Course Title: Research Project /Dissertation	
Max Marks: 100	Core: Compulsory	
Course Outcomes(COs)		
9. Develop the ability to conduct independent research in specific area of mathematics. 10.Enhance critical thinking skills by analysing and evaluating existing literature and experimental data. 11.Demonstrate the capacity to solve complex mathematical problems related to the chosen topic. 12.Improve scientific writing skills, including the ability to clearly and concisely communicate research findings. 13.Gain proficiency in data analysis techniques and the use of relevant software and techniques. 14.Present research findings through oral presentations and written reports. 15.Make a unique contribution to the field by adding new insights, findings, or experimental data. 16.Understand and adhere to ethical standards in research, including proper citation and plagiarism avoidance.		
Note: 1. Every student of third year has to complete a Research Project/Dissertation which will be allotted in the 5th semester. 2. This Research Project/Dissertation may be Interdisciplinary. 3. The Research Project/Dissertation will be done under supervision of a teacher. If needed, a co-supervisor may be allotted from any Industry/ University/ Technical Institute/ Research Institute etc. 4. The student will submit a joint report of research project/Dissertation done in both semesters (5th and 6th semesters) at the end of the year (end of 6th semester), which will be evaluated jointly by the supervisor and the external examiner appointed by the college on total marks 100.		